

ECE 310: Digital Signals Processing, Spring 2025

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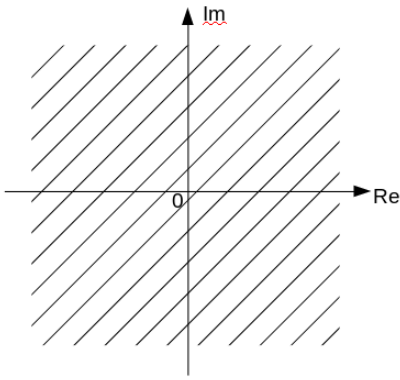
**Solution:**

$$\begin{aligned} X(z) &= \sum_{n=-\infty}^{\infty} x[n]z^{-n} \\ &= \sum_{n=-\infty}^{\infty} (\delta[n+1] - 2\delta[n-2])z^{-n} \\ &= z - 2z^{-2}, \text{ ROC: } z \neq 0, |z| < \infty \end{aligned}$$

A diagram of the complex plane with a horizontal axis labeled 'Re' and a vertical axis labeled 'Im'. The origin is marked with '0'. A region is shaded with diagonal lines, representing the set of complex numbers  $z$  such that  $\operatorname{Re}(z) < 0$  and  $\operatorname{Im}(z) > 0$ .

$$\begin{aligned} X(z) &= \sum_{n=-\infty}^{\infty} x[n]z^{-n} \\ &= \sum_{n=-\infty}^{\infty} (-\delta[n+1] + 2\delta[n] + 3\delta[n-2] + \delta[n-3])z^{-n} \\ &= \boxed{-z + 2 + 3z^{-2} + z^{-3}, \text{ ROC: } z \neq 0, |z| < \infty} \end{aligned}$$

ROC:



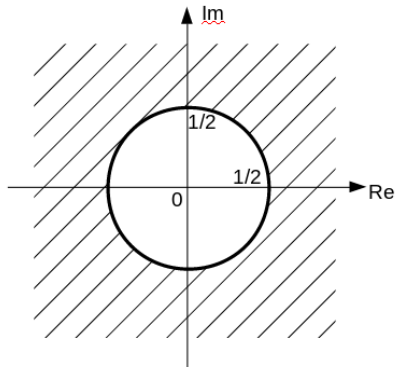
c)  $x[n] = \left(\frac{1}{2}\right)^{n-1} u[n-2]$

$$\begin{aligned} X(z) &= \sum_{n=-\infty}^{\infty} x[n]z^{-n} \\ &= \sum_{n=-\infty}^{\infty} \left(\frac{1}{2}\right)^{n-1} u[n-2]z^{-n} \\ &= 2 \sum_{n=-\infty}^{\infty} \left(\frac{1}{2}\right)^n u[n-2](z^{-1})^n \\ &= 2 \sum_{n=2}^{\infty} \left(\frac{1}{2}z^{-1}\right)^n \end{aligned}$$

here use the identity  $\sum_{k=m}^{\infty} r^k = \frac{r^m}{1-r}$ ,  $|r| < 1$ .

$$\begin{aligned} H(z) &= 2 \frac{\left(\frac{1}{2}z^{-1}\right)^2}{1 - \frac{1}{2}z^{-1}}, \quad \left|\frac{1}{2}z^{-1}\right| < 1 \\ &= \frac{1}{2z^2 - z}, \quad \text{ROC: } |z| > \frac{1}{2} \end{aligned}$$

ROC:



**Alternative Solution:** Using the z-transform pair

$$x_1[n] = a^n u[n] \leftrightarrow X_1(z) = 1/(1 - az^{-1}), \quad \text{ROC: } |z| > |a|$$

and combine with the shifting and scaling properties:

$$x_2[n] = \beta x_1[n - k] \leftrightarrow X_2(z) = \beta z^{-k} X_1(z), \text{ ROC: } R_{X_1} \text{ except } z = 0 \text{ or } \infty.$$

Now, let  $x_1[n] = (\frac{1}{2})^n u[n]$ . Then,  $x[n] = \frac{1}{2} x_1[n - 2]$ . Hence,

$$X_1(z) = \frac{1}{1 - \frac{1}{2}z^{-1}}, \text{ ROC: } |z| > \frac{1}{2}.$$

$$\begin{aligned} X(z) &= \frac{1}{2} z^{-2} X_1(z), \text{ ROC: } |z| > \frac{1}{2} \\ &= \boxed{\frac{1}{2} z^{-2} \frac{1}{1 - \frac{1}{2}z^{-1}}, \text{ ROC: } |z| > \frac{1}{2}} \end{aligned}$$

**d)**  $x[n] = 5(3)^n u[-n + 2] + 3\left(\frac{1}{3}\right)^{n-3} u[n + 3]$

We can take the z-transforms of the two terms separately and sum the results. The ROC we will be the intersection of the ROCs.

Let  $x_1[n] = 5(3)^n u[-n + 2]$  and  $x_2[n] = 3\left(\frac{1}{3}\right)^{n-3} u[n + 3]$  where  $x[n] = x_1[n] + x_2[n]$ .

Also let  $x_0[n] = \left(\frac{1}{3}\right)^n u[n]$ . Then  $x_1[n]$  and  $x_2[n]$  can be rewritten in terms of  $x_0[n]$  as:

$$\begin{aligned} x_1[n] &= 5(3)^2 x_0[-n + 2] \\ x_2[n] &= 3^7 x_0[n + 3] \end{aligned}$$

The z-transform of  $x_0[n]$  is simply:

$$X_0(z) = \frac{1}{1 - \frac{1}{3}z^{-1}}, \text{ ROC: } |z| > \frac{1}{3}.$$

Now we can use the time shifting, scaling, and time reversal properties of the z-transform to get  $X_1(z)$  and  $X_2(z)$ .

Define  $\hat{x}_1[n] = 5(3)^2 x_0[n + 2]$ , hence,  $x_1[n] = \hat{x}_1[-n]$ . Hence,

$$\hat{X}_1(z) = 5(3)^2 z^2 X_0(z), \text{ ROC: } |z| > \frac{1}{3}$$

$$\begin{aligned} X_1(z) &= \hat{X}_1(z^{-1}), \text{ ROC: } |z^{-1}| > \frac{1}{3} \\ &= 5(3)^2 z^{-2} X_0(z^{-1}), \text{ ROC: } |z| < 3 \\ &= \frac{45z^{-2}}{1 - \frac{1}{3}z}, \text{ ROC: } |z| < 3, |z| \neq 0 \end{aligned}$$

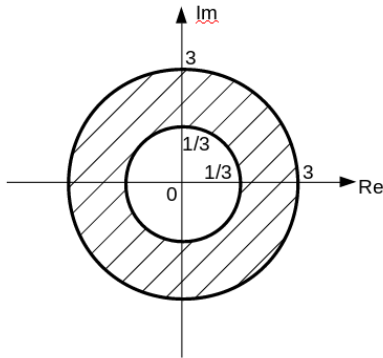
and

$$\begin{aligned} X_2(z) &= 3^7 z^3 X_0(z), \text{ ROC: } |z| > \frac{1}{3}, |z| < \infty \\ &= \frac{3^7 z^3}{1 - \frac{1}{3}z^{-1}}, \text{ ROC: } |z| > \frac{1}{3}, |z| < \infty \end{aligned}$$

Combining  $X_1(z)$  and  $X_2(z)$ :

$$X(z) = \frac{45z^{-2}}{1 - \frac{1}{3}z} + \frac{3^7 z^3}{1 - \frac{1}{3}z^{-1}}, \text{ ROC : } \frac{1}{3} < |z| < 3$$

ROC:



**Grading:** 21 points

- +3 expression parts a-c
- +2 ROC parts a-c
- +4 expression part d
- +2 ROC part d

2

**Solution:**

a)

$$y[n] = x[n - 2]$$

Using the time shift property:

$$Y(z) = X(z)z^{-2} = \frac{z^{-2}}{1 - \frac{1}{2}z^{-1}}, \text{ ROC: } |z| > \frac{1}{2}$$

b)

$$y[n] = n^2 x[n]$$

This can be seen as the differentiation in frequency property applied twice. First, an intermediate signal is defined as:

$$x_1[n] = nx[n]$$

$$X_1(z) = -z \frac{dX(z)}{dz}$$

$$X_1(z) = -z \left( -\frac{2}{(2z - 1)^2} \right)$$

$$X_1(z) = \frac{2z}{(2z - 1)^2}$$

$$Y(z) = -z \frac{dX_1(z)}{dz}$$

$$Y(z) = \boxed{\frac{2z(2z+1)}{(2z-1)^3}, \text{ROC: } |z| > \frac{1}{2}}$$

c)

$$y[n] = 3^n x[n]$$

$$Y(z) = X(3^{-1}z)$$

$$Y(z) = \frac{1}{1 - \frac{1}{2}(\frac{z}{3})^{-1}}$$

$$Y(z) = \boxed{\frac{1}{1 - \frac{3}{2}z^{-1}}, \text{ROC: } |z| > \frac{3}{2}}$$

d)

$$y[n] = \cos(\frac{\pi}{4}n)x[n]$$

$$y[n] = \frac{1}{2}(e^{j\frac{\pi}{4}n} + e^{-j\frac{\pi}{4}n})x[n]$$

$$y[n] = \frac{1}{2}e^{j\frac{\pi}{4}n}x[n] + \frac{1}{2}e^{-j\frac{\pi}{4}n}x[n]$$

$$Y(z) = \frac{1}{2}X(e^{-j\frac{\pi}{4}}z) + \frac{1}{2}X(e^{j\frac{\pi}{4}}z)$$

$$Y(z) = \boxed{\frac{1}{2} \frac{1}{1 - \frac{1}{2}(e^{-j\frac{\pi}{4}}z)^{-1}} + \frac{1}{2} \frac{1}{1 - \frac{1}{2}(e^{j\frac{\pi}{4}}z)^{-1}}, \text{ROC: } |z| > \frac{1}{2}}$$

e)

$$y[n] = (x * u)[n]$$

Using the convolution property:

$$Y(z) = X(z) \frac{1}{1 - z^{-1}}$$

$$Y(z) = \boxed{\frac{1}{1 - \frac{1}{2}z^{-1}} \frac{1}{1 - z^{-1}}, \text{ROC: } |z| > 1}$$

f)

$$y[n] = (x * h)[n], h[n] = \delta[n] - \frac{1}{2}\delta[n-1]$$

Taking the Z-Transform of  $h[n]$ :

$$H(z) = 1 - \frac{1}{2}z^{-1}$$

Using convolution properties:

$$Y(z) = X(z)H(z)$$

$$Y(z) = \frac{1}{1 - \frac{1}{2}z^{-1}}(1 - \frac{1}{2}z^{-1})$$

$$Y(z) = \boxed{1, \text{ROC: } |z| \in \mathcal{R}}$$

**Grading:** 30 points

- +4 each part expression
- +1 each part ROC

**Solution:**

a)

$$x[n] = \left(\frac{1}{2}\right)^{n-1} \sin\left(n\frac{\pi}{4} + \frac{\pi}{3}\right) u[n-2]$$

To get a common delay in n:

$$x[n] = \left(\frac{1}{2}\right)\left(\frac{1}{2}\right)^{n-2} \sin\left(\frac{(n-2)\pi}{4} + \frac{5\pi}{6}\right) u[n-2]$$

Considering the general form:

$$y[n] = a^n \sin(\omega_o n + \phi) u[n]$$

Using Euler to expand:

$$y[n] = \frac{1}{2j} a^n (e^{j\omega_o n} e^{j\phi} - e^{-j\omega_o n} e^{-j\phi}) u[n]$$

$$y[n] = \frac{1}{2j} a^n e^{j\omega_o n} e^{j\phi} u[n] - \frac{1}{2j} a^n e^{-j\omega_o n} e^{-j\phi} u[n]$$

$$Y(z) = \left(\frac{1}{2j} e^{j\phi}\right) \frac{1}{1 - az^{-1} e^{j\omega_o}} - \left(\frac{1}{2j} e^{-j\phi}\right) \frac{1}{1 - az^{-1} e^{-j\omega_o}}$$

$$Y(z) = \frac{\frac{1}{2j} z^2 (e^{j\phi} - e^{-j\phi}) + \frac{1}{2j} az (e^{j(\omega_o - \phi)} - e^{-j(\omega_o - \phi)})}{z^2 - az(e^{j\omega_o} + e^{-j\omega_o}) + a^2}$$

$$Y(z) = \frac{z^2 \sin(\phi) + az \sin(\omega_o - \phi)}{z^2 - 2az \cos(\omega_o) + a^2}, \text{ROC: } |z| > |a|$$

$$X(z) = \frac{1}{2} \frac{z^2 \sin(\frac{5\pi}{6}) + (\frac{1}{2}) z \sin(\frac{\pi}{4} - \frac{5\pi}{6})}{z^2 - 2\frac{1}{2} z \cos(\frac{\pi}{4}) + (\frac{1}{2})^2} z^{-2}$$

$$X(z) = \frac{1}{2} \frac{\frac{1}{2} z^2 + \frac{1}{2} z(-0.966)}{z^2 - \frac{z}{\sqrt{2}} + \frac{1}{4}} z^{-2}$$

$$X(z) = \boxed{\frac{1}{2} \frac{\frac{1}{2} + \frac{1}{2} z^{-1}(-0.966)}{z^2 - \frac{z}{\sqrt{2}} + \frac{1}{4}}, \text{ROC: } |z| > \left|\frac{1}{2}\right|}$$

b)

$$x[n] = 2^n u[n] + \left(\frac{1}{2}\right)^n u[n-2]$$

$$x[n] = 2^n u[n] + \frac{1}{4} \left(\frac{1}{2}\right)^{n-2} u[n-2]$$

Using transform pairs:

$$X(z) = \boxed{\frac{1}{1 - 2z^{-1}} + \frac{1}{4} \frac{z^{-2}}{1 - \frac{1}{2}z^{-1}}, \text{ROC: } |z| > |2|}$$

c)

$$x[n] = 2^n (u[n] - u[n-8])$$

$$x[n] = 2^n (\delta[n] + \delta[n-1] + \delta[n-2] + \delta[n-3] + \delta[n-4] + \delta[n-5] + \delta[n-6] + \delta[n-7])$$

$$x[n] = \delta[n] + 2\delta[n-1] + 2^2\delta[n-2] + 2^3\delta[n-3] + 2^4\delta[n-4] + 2^5\delta[n-5] + 2^6\delta[n-6] + 2^7\delta[n-7]$$

Using transform pairs:

$$X(z) = 1 + 2z^{-1} + 2^2z^{-2} + 2^3z^{-3} + 2^4z^{-4} + 2^5z^{-5} + 2^6z^{-6} + 2^7z^{-7}, \text{ROC: } |z| \neq 0$$

d)

$$x[n] = n\left(\frac{1}{2}\right)^n u[n-3]$$

$$x[n] = n\left(\frac{1}{8}\right)\left(\frac{1}{2}\right)^{n-3} u[n-3]$$

$$x[n] = (n-3)\left(\frac{1}{8}\right)\left(\frac{1}{2}\right)^{n-3} u[n-3] + \frac{3}{8}\left(\frac{1}{2}\right)^{n-3} u[n-3]$$

Using transform pairs:

$$X(z) = \frac{1}{8} \frac{\frac{1}{2}z^{-1}}{\left(1 - \frac{1}{2}z^{-1}\right)^2} z^{-3} + \frac{3}{8} \frac{z^{-3}}{1 - \frac{1}{2}z^{-1}}, \text{ROC: } |z| > \frac{1}{2}$$

$$X(z) = \frac{1}{16} \frac{z^{-4}}{\left(1 - \frac{1}{2}z^{-1}\right)^2} + \frac{3}{8} \frac{z^{-3}}{1 - \frac{1}{2}z^{-1}}, \text{ROC: } |z| > \frac{1}{2}$$

**Grading:** 28 points

- +5 expression each part
- +2 ROC each part

4

### Solution:

a)

$$X(z) = \frac{z^2 - z}{z^2 + 3z + 2} = \frac{z(z-1)}{(z+1)(z+2)} = \frac{-2z}{z+1} + \frac{3z}{z+2}$$

$$x[n] = 3(-2)^n u[n] - 2(-1)^n u[n] \quad \text{For ROC: } |z| > 2$$

$$x[n] = -3(-2)^n u[-n-1] - 2(-1)^n u[n] \quad \text{For ROC: } 1 < |z| < 2$$

$$x[n] = -3(-2)^n u[-n-1] + 2(-1)^n u[-n-1] \quad \text{For ROC: } 0 \leq |z| < 1$$

b):

$$X(z) = \frac{z^3 - z}{z^2 + 3z + 2} = \frac{z(z^2 - 1)}{(z+1)(z+2)} = \frac{z(z+1)(z-1)}{(z+1)(z+2)} = \frac{z^2 - z}{(z+2)}$$

$$x[n] = (-2)^{(n+1)} u[n+1] - (-2)^n u[n] \quad \text{For ROC: } 2 < |z| < \infty$$

$$x[n] = -(-2)^{(n+1)} u[-n-2] + (-2)^n u[-n-1] \quad \text{For ROC: } 0 \leq |z| < 2$$

c):

$$X(z) = \frac{\frac{5}{2}}{1 - \frac{1}{3}z^{-1}} - \frac{\frac{3}{2}}{1 - \frac{1}{5}z^{-1}}$$

$$x[n] = \boxed{\frac{5}{2} \left(\frac{1}{3}\right)^n u[n] - \frac{3}{2} \left(\frac{1}{5}\right)^n u[n] \quad \text{For ROC: } |z| > \frac{1}{3}}$$

$$x[n] = \boxed{-\frac{5}{2} \left(\frac{1}{3}\right)^n u[-n-1] - \frac{3}{2} \left(\frac{1}{5}\right)^n u[n] \quad \text{For ROC: } \frac{1}{5} < |z| < \frac{1}{3}}$$

$$x[n] = \boxed{-\frac{5}{2} \left(\frac{1}{3}\right)^n u[-n-1] + \frac{3}{2} \left(\frac{1}{5}\right)^n u[-n-1] \quad \text{For ROC: } 0 \leq |z| < \frac{1}{5}}$$

**Grading:** 21 points

- +8 for (a) correct possible ROC and corresponding inverse z-transform
- +5 for (b) correct possible ROC and corresponding inverse z-transform
- +8 for (c) correct possible ROC and corresponding inverse z-transform